

Master Regulators for Metastatic behavior in Osteosarcoma

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Osteosarcomas mainly affect young people, and are fatal in 80% of metastatic cases. Some genes seem to be more involved than others in the development of metastases. To study this phenomenon, we developed an inferred transcriptional regulatory network from a microarray of 53 patients. The main goal is to identify the transcriptional factors that would be responsible for the development of these metastases. Indeed, these super-regulators are responsible for signalisation cascades, some of which are identified as being involved in cellular development and proliferation. It appears that the expression of the HOXC4, VENTX, IRF8, VAV1 and SP140 genes is strongly correlated with the absence of metastases. HOXC4 and VENTX are homeobox genes, with VENTX being a suppressor of tumor, while IRF8, VAV1 and SP140 are involved in the inflammatory responses of the immune system. These new inference algorithms could make it possible to determine the origin of certain pathologies far upstream of the signalisation cascades.

Osteosarcoma | Transcriptional regulatory network | Microarray | Transcription Factor | Master Regulator genes | Differential gene analysis

Introduction

Osteosarcoma or osteogenic sarcoma (OGS) is a tumoral cancer localized in the bone. This pathology affects approximately 42% of women and 58% of men (1). Osteosarcoma occurs differently according to ethnicity, with highest incidence rate in Blacks, followed by Hispanics and Caucasians respectively (1). This type of musculoskeletal cancer mainly affects young people and is one of the five most common cancers in children and adolescents (2). Concerning the epidemiology of osteosarcoma, it is estimated that 42% of primary tumors are located in the femur (75% in the distal femur), 19% located in the tibia (with 80% in the proximal part) and 10% in the humerus (with 90% in the proximal part) (1). Patients can develop metastasis, a group of tumoral cells at distance from the primary tumor site. The metastasis first detaches from the primary site, going through the circulatory and lymphatic systems and aggregates in a secondary site (3). The number of patients with metastases evidences is estimated at 73% (4). Moreover metastatic osteosarcoma is often fatal, with 80% of patients with metastases dying (5). There are multiple histological subtypes of this cancer including chondroblastic, osteoblastic, fibroblastic, sclerosing (6) (7). There are many hypotheses concerning the development of metastases and there is no scientific consensus at the moment. One of these hypothesis is a development issue in epithelial-mesenchymal transitions (8). Another hypothesis, more specific to bone cancer, is a defective osteogenic differentiation (9).

Stromal cells, a type of cell with stem cell abilities that originates from bone marrow seem to be implicated in the development of bone cancer, with an osteoblastic differentiation

(10). Communication between stroma and tumor cells seems to be a promising area of research on the anarchic division of cancer cells (11). Mutations in stromal cell are correlated with different kinds of cancer, including osteosarcoma (12) (13)(14).

It therefore seems essential to study these genetic factors. Next generation sequencing enables to study the whole genome of patients in order to find some genetic factors implicated in some pathologies (15). Illumina sequencing is one of these next generation sequencing techniques, which enables transcriptomic quantification of organisms, including mammals (16).

With the increasing amount of data to processed, it is now necessary to develop high level analytical techniques (17). In light of the foregoing, trying to reproduce biological networks seems to be one of the new challenges ahead (18). Indeed, our genes do not work alone and are regulated by each other. Regulatory networks provide an overview of the complexity of life and allow us to better understand how it works. The usage of regulatory networks is therefore an innovative approach that can enable us to understand diseases from a more integrated point of view (19) and can enable to discover new regulator of genetic risks(20). In this project, we are studying the link between tumor cells, their stromal environment and the development of metastases in osteosarcoma.

Buddingh et al. (2011) worked on identifying the link between tumor cells, their stromal environment and the development of metastasis in osteosarcoma and generated a MIAME (minimum information about a microarray experiment) compliant data, published in the GEO database (www.ncbi.nlm.nih.gov/geo/, accession number GSE21257) (5). This GEO set contains tumor cells and development of metastasis data for 53 patients with osteosarcoma (who developed or not metastasis within 5yrs). Here, instead of focusing on individual genes, we endeavored to construct a regulatory network of genes to identify genes differentially expressed in metastases and in particularly Master Regulator (MR) genes. These MRs were first hypothesized by Ohno in 1930. They are up-stream of the regulation hierarchy and, as such, regulate other genes's signalisation cascades (21). As such, this allows us to identify regulatory genes of interest and investigate more deeply all the signalisation cascades they are implicated in (22).

Materials and Methods

Acquisition of the GEO set.

Author contributions:

G.F., K.K. and V.S. contributed equally to this work.

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Table 1. Different subtypes of osteosarcoma

Subtype	Cohorts in %
Osteoblastic	60.37
Chondroblastic	11.32
Fibroblastic	9.43
Telangiectatic	5.66
Sclerosing	3.77
Others	9.44

Bone cancer is divided into several types. The majority being osteoblastic followed by chondroblastic, fibroblastic, sclerosing, telangiectatic and finally other less common types.

This part of the experiment was done by Buddingh et al (2011). They published in the GEO database the MIAME (minimum information about a microarray experiment) compliant data (www.ncbi.nlm.nih.gov/geo/, accession number GSE21257) that we used for our analysis.

Patient cohorts. The Genome-wide expression profiling was performed on pretreated and snap-frozen diagnostic biopsies from 53 resectable osteosarcoma patients of the Centre for Molecular Medicine, Norway. 56,6% of these patients were alive after diagnosis. The mean age of this cohort is 224,55 ($\pm 146,35$) months. There are 35,85% of females and 64.15% of males. Tumors are located in different parts of the body, with 50.9% of tumors occurring on the femur, 28.3% on the tibia, 15.1% on the humerus, 3.8% on the fibula and the rest being unknown. This is consistent with the epidemiological data (1). There are multiple subtypes of osteosarcoma, the most prevalent is the osteoblastic subtype, followed by chondroblastic, fibroblastic and telangioblastic (Table 1). Finally, there are around 35,85% of patients that had no metastasis at the time of the biopsies.

RNA isolation, cDNA synthesis, cRNA amplification, and Illumina Human-6 v2.0 Expression BeadChip hybridization. Biopsy tissues were sampled, stored at -70°C and cut into $20\mu\text{m}$ slices by cryostat. The RNA was isolated with TRIzol, purified and RNA quality was controlled. An RT-PCR was realised from RNA to obtain cDNA. Finally, cDNA synthesis, cRNA amplification and cRNA hybridization were performed with Illumina Human-6 v2.0 Expression BeadChip. Data were normalized through usage of 3 reference genes (Fig. 1).

Microarray data analysis. Gene expression data were imported using BeadStudio version 3.1.1.3.0 (Illumina) in a GeneSpring probe format, to be then processed and analyzed with R software. The MIAME (minimum information about a microarray experiment) compliant data obtained was then exported to the GEO database (www.ncbi.nlm.nih.gov/geo/, accession number GSE21257) by Buddingh et al (2011).

MIAME analysis.

We downloaded for our analysis GSE21257 GEO set, acquired by Buddingh et al. (2011), and used the R software for our analysis. We choose to avoid filtering the data, since the aim of the project is to study the regulatory network. Filtering could strongly skew the results by removing, for instance, a regulator gene with slight variations. Additionally, the data were previously normalized by Buddingh et al.

GSE21257/GPL10295 selected sample

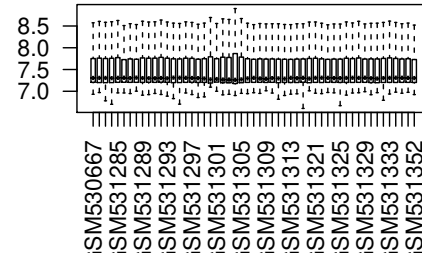


Fig. 1. Boxplot for selected GEO samples : The data was normalized, as can be seen in the sample gene expression selections above (5).

(2011). GSE21257 was in a S4 format containing a microarray matrix (named "ExpData" in the GSE21257 GEO set provided) corresponding to the 53 samples described previously (1 for each of the from 53 patients) and the 48701 probes brought to light by the microarray. We have a file in S4 format with inside the microarray matrix (called ExpData in the script provided) with 53 samples (from 53 patients) and 48701 probes (complementary sequences to target mRNAs used in microarray to evaluate gene expression). Other information is available in the dataset, including information about pre-chemotherapy tissues, patients and so on.

The probes ("ID") having been used to identify gene expression, we added to the matrix the names of the genes ("SYMBOL") that where hybridized to the probes. Probes that couldn't be associated with the gene they label (about 37%) were ignored for the rest of the project. It is important to note that multiple probes can label the same gene. In such a case, usage of the R package "RTN" allowed us to choose the most informative probes (by default) through variation coefficients.

Inference of the regulatory network.

So as to visualize our data, we created a TNI-class object to infer a regulatory network. To do so, we used the RTN R package. RTN stands for Reconstruction of Transcriptional regulatory Networks and analysis of regulons or shortly RTN and makes it possible to collect transcription factors (TFs) and their target genes. A regulon is a section of prokaryotic DNA that contains one or more genes controlled by the same transcription factor. Before plotting the RTN graph, a pre-process is required to infer the network on the basis of all known TFs. After that, to remove the weakest interactions (inferred by mutual information) 1000 permutations were implemented. The "snow" R package was used to speed up the permutation process. After that, bootstrap analysis with 1000 iterations (random iterative process) allowed us to eliminate unstable interactions. The DPI (Data Processing Inequality) filter function then allowed us to use "ARACNe" (Algorithm for the Reconstruction of Accurate Cellular Networks) so as to eliminate possible indirect interactions by computing all pairs with mutual information and by eliminating links below a certain threshold (23). All graphs were obtained through usage of the RedR R package.

Differential gene expression analysis.

We used gene expression analysis obtained by Buddingh et al. (2011) (GSE21257 GEO set) and the [limma](#) R package to compute differential gene expression between metastatic and non-metastatic biopsies (binary phenotype). The 250 most significantly differentially expressed genes (or "hits") were then selected to act as "signature" of osteosarcoma metastatic behavior.

Master Regulators analysis.

To visualize which genes regulate others the most, we computed a Master Regulator analysis through usage of the RTN package. This requires a preprocessing of previously obtained TNI-class object into a TNA-class object (TNA : Transcriptional Network Analysis). To do so, we used the hits list obtained from the differential gene expression analysis and the TNI-class object obtained from the regulatory network inference. Since the transcriptional networks are by nature scale-free and have a huge number of both nodes and edges (24), it is therefore very difficult to interpret. As such, the MR approach allows us to refine the visualised network by focusing on signature transcription factors, in order to simplify interpretation.

One-Tailed Gene Set Enrichment Analysis (GSEA1).

The gene set enrichment analysis (GSEA1) was implemented through usage of the RTN package (with 1000 permutations). GSEA1 uses previously generated TNA-class object to test association between signature MRs and patient condition, more specifically by using scores obtained during differential gene expression analysis. As such, GSEA1 shows how strongly signature MRs regulate other genes throughout generation of enrichment scores.

Two-tailed GSEA (GSEA2).

We used RTN (with 1000 permutations) to compute GSEA2 from TNA-class object previously obtained. This enrichment analysis is based on the CMAP procedure (Connectivity map procedure) and gives us more details concerning direction of regulations by specific signature MRs. GSEA2 uses Pearson's correlation to divide into 2 subgroups each regulon, one containing positive targets and the other negative targets (obtained from Pearson's correlation (positive or negative)). GSEA2 also ranks genes, using the logFC phenotype scores (represents magnitude of a certain phenotype) obtained during differential gene expression analysis, from 2 (representing presence of metastasis) to -2 (representing absence of metastasis). This allows GSEA2 to generate distinct enrichment scores for each subgroup. As such, GSEA2 allows us to evaluate differential enrichment between each subgroup and determine relation between phenotype and regulon activity (up- or down- regulated).

Results

Inferred transcriptional network implies in osteosarcomatic metastasis.

A histogram of nodes degree was done from the TNI-class object generated (**Fig. 2**). This histogram shows that most of the nodes in our network are regulated by less than 10 genes. We can even add that the majority of them are regulated by only one gene.

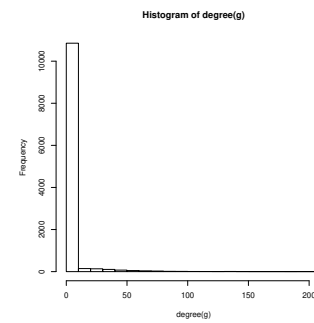


Fig. 2. Histogramm of nodes degree : Node degree histogram shows that the most majority of nodes are regulated by less than 10 TFs

We obtained the transcriptional network (TN) from known transcription factors (TFs). Our network contains 11489 TFs and 25 287 edges, represented throughout a dendrogram (amapDend) (**Fig. 3**). The Master Regulator analysis highlighted (in red) the 64 most significant MRs (adj p-value < 0.05). The closest MRs are to each other, the more they regulate common genes. The size of the nodes represents the size of the regulons, also known as the degree (number of edge connected to that node). None of the nodes seems to be particularly isolated.

Master Regulators and enrichment analysis.

Master Regulator analysis identifies 570 MRs, with 64 of them being highly enriched. The association map (**Fig. 4**) represents the similarity degree among interfered regulons. The width of edges depends of the Jaccard coefficient (JC). A thick edge corresponds to a high similarity between interferons. Circles are represented in different colors, from blue (low adjusted p-value) for highly correlation between gene and phenotype, to red (high adjusted p-value) for low correlation between gene and phenotype. A grey circle represents the absence of correlation with phenotype (adj pval around 0.5). The size of the circle represents the degree (number of connections between the TF and its targets). Isolated nodes are eliminated. RXRG and ANKRD6, for example, are both isolated from the rest of the network and have poorly significant ES.

Top 5 MRs.

We choose to focus on the top 5 MRs with their targets which are HOXC4, IRF8, SP140, VAV1 and VENTX.

The **Fig. 5** shows the get set enrichment analysis focused on the top 5 MRs. The upper part of the figure represent the ranked phenotype in y-axis from 0 (non metastatic phenotypes) to 2 (metastatic phenotypes). The middle graph represent the regulon (genes regulate by the same MR) for each top 5 MRs. And the lower part represent the ES from 0 (not significantly different for a given phenotype) to 0.8 (significantly different accordint patient's conditions). This graph shows that genes with the highest absolute logFC value (ranked in the firsts positions) are the most linked with a certain patient's condition. With a high absolute logFC value, there are a huge number of expressed genes in the regulon. Finally, for the highest absolute logFC ranked value, the absolute magnitude of the phenotype tends to be around 2.

The regulatory network shows the transcription factor (TF)-target interactions of the five master regulators, each

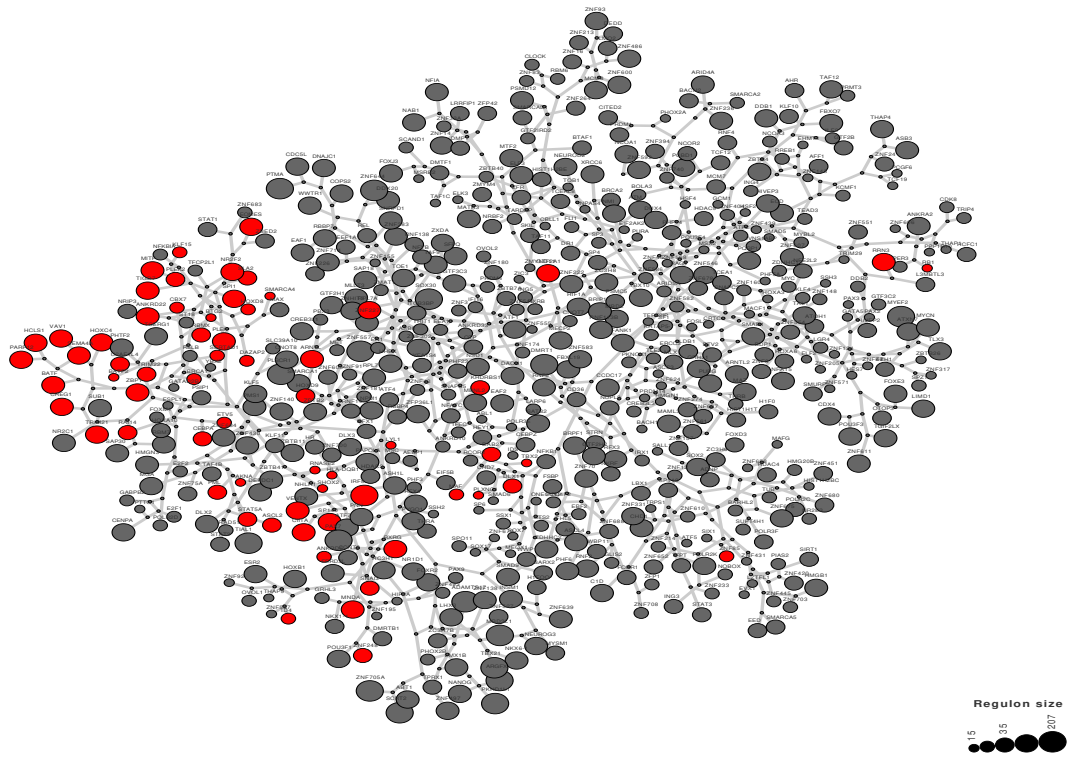


Fig. 3. Tree and leaf representation of the network : The dendrogram (amapDend) show the transcriptional network with nodes represent the TFs and branches, the arms of the dendrogram. The size of nodes represents the number of genes regulated by the same TFs. MRs are represented in red. The closest MRs regulate the same genes.)

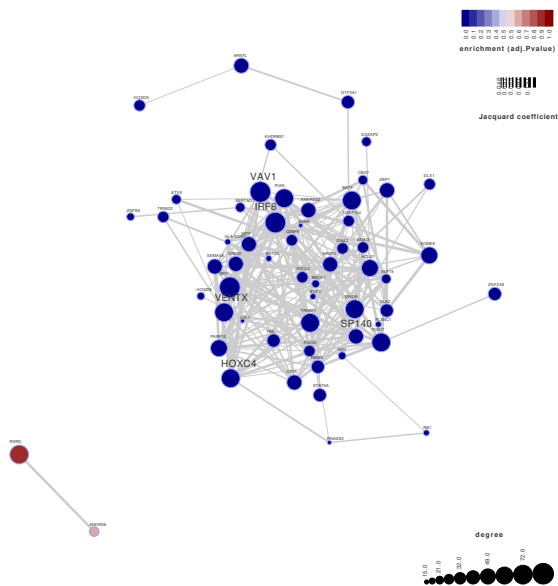


Fig. 4. Association map of the inferred TN The MRs association map use the Jaccard coefficient to compute the similarity degree among inferred regulons. The size of the nodes represent the degree that are colored from blue (adj p-val < 0.05 for enrichment score) to red (high adj p-val)

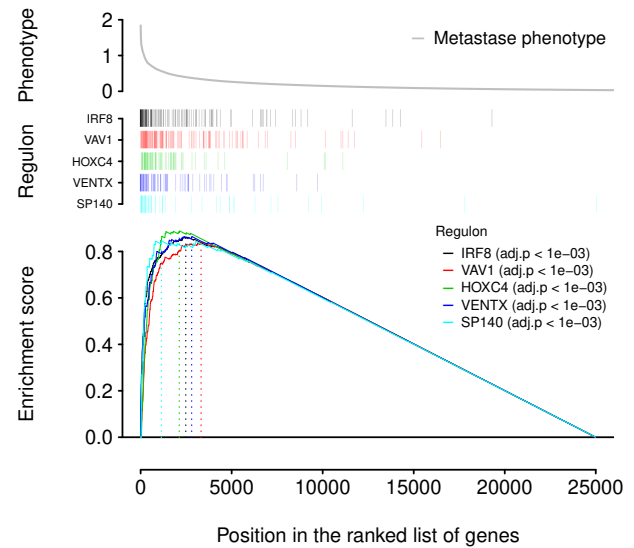


Fig. 5. Gene set enrichment plot of the top 5 MRs according to phenotypes : The x-axis represent the list of gene ranked from 0 to 25 000 by their respective absolute logFC. The upper part of the graph represent in y-axis the phenotypes ranked via the magnitude of a certain phenotypes (logFC) from 0 (no linked to metastasis) to 2 (linked to absence or presence of metastasis). The middle part of the graph represent the regulons for each top 5 MRs. The lower part of the graph represent the ES for each ranked genes

one containing one TF (square nodes) and all inferred targets (round nodes). The mode of action represented in red/blue colors corresponds to the correlation pattern observed between a given transcription factor and its targets, assessed by the Pearson's correlation on TN1.

The transcriptional network illustrated in **figure 6** is a filtered one, that focus on the TFs targets (little square) of the top 5 MRs (big square) find previously. There are 342 targets and 468 edges. This graph allows to see the action mode of these regulators, with a red link for an activation and a blue one for an inhibition. Some targets are both activated and inhibited by different genes. IRF8, VAV1, VENTX, HOXC4 and SP140 regulates respectively 143, 130, 75, 71 and 49 TFs.

The 2-tailed GSEA **Fig. 7** is done for each choosen MRs. The highest variables gens are ranked in x-axis from 0 (logFC near to 2) to 25 000 (logFC near to -2). The phenotypes goes from -2 (no metastasis) to 2 (metastasis). The ES is shown for activated genes (red) and inhibited ones (blue). HOXC4 inhibit all his targets. The lowest ES of the targets correspond to non metastatic phenotypes. For IRF8, ES is high for inhibited targets with a phenotype that tend to metastatic, and low for positive targets with a non metastatic condition. Concerning SP140, it seems to activate all his targets. ES magnitude is higher with a negative ES for positive target with a non metastatic condition. For VAV1 we have a high magnitive of ES for negative targets with a positive ES for metastatic condition. We have also a high negative ES for positive targets with non-metastatic condition. Finally for VENTX there are high positive ES for negative target in metastatic position and high negative ES for positive target with non-metastatic condition.

Discussion.

Our main goal was to identify regulatory units most likely to be implicated in the presence of metastasis in osteosarcoma. For that purpose, we inferred from a gene expression data set originating from osteosarcoma patients (with or without metastasis) a regulatory transcriptional network.

The nodes degree histogram (**Fig. 2**) confirms that TNs are free-scale network (24). Indeed, most of the nodes (TFs) are linked to less than 10 other nodes. We assume that this confirms the non-aleatory nature of our network since the a small number of "hubs" are present. A hub (or MR) is described as a node with more than 10 interactions.

The dendrogram representation of our network (**Fig. 3**) illustrates that the MRs, identified by MR analysis, are scattered all over the network. Most of them are close to each other. This allows us to hypothesize that they share a lot of information and therefore share a lot of common targets. This is confirmed by the TN association map visualization, that shows all most significant MRs **Fig. 4**. Indeed, a large fraction of the MRs are linked to each other. This representation also allows us to notice that these MRs are also the ones that are significantly implicated in the metastasis phenotype, according to enrichment scores (**Fig. 4**). We focused our analysis on the 5 MRs most significantly associated with the phenotype : IRF8, VAV1, HOXC4, VENTX and SP140. These genes's GSEA1 enrichment scores reveal that they are strongly linked to the studied phenotype.

IRF8 belongs to the interferon regulatory factor (IFN) family. This gene family binds to the IFN-simulated response element, controls the expression of IFN-alpha and

IFN-beta caused by viral infection, and is implicated in regulation of genes stimulated by these two genes according to [IRF8 gene card](#). According to [VAV1 gene](#) VAV1 is a guanine nucleotide exchange factors which encode a protein important for the development and activation of T-cell and B-cell. Protein encoded by VENTX, member of VENT family of homeodomain proteins, are important in hematopoietic stem cell maintenance and in mesodermal patterning and work as a transcriptional repressor as stated by [VENTX gene card](#). SP140 protein, member of the SP100 family, is interferon-inductible, a hight levels of this protein are found in the nuclei of leukocytes according to [SP140 gene](#) Stated by [HOXC4 gene](#) HOXC4 is a gene, belongs to the homeobox family, which encode transcription factors implicated in morphogenesis in multicellular organisms. He is located in chromosome 12.

From our inferred network, we can hypothesize that when HOXC4 is expressed, genes implicated in the expression of the non-metastasis phenotype are inhibited. As such, we can conclude that HOXC4 could be implicated in the development of metastases in osteosarcoma. Indeed, it's a HOX genes that is implicated in cell differentiation and proliferation (25). Contrariwise, expression of IRF8, VAV1, VENTX and SP140 in our network seems to induce expression of non-metastasis related genes. More specifically, IRF8 plays a role in osteosarcoma by upregulating transforming growth factor-beta type I receptor (TGF-B)) (26). On the other hand VENTX is identified as a suppressor of tumor. This gene code a protein, VentX responsive of p53 apoptosis (27). SP140 seems to be regulated by interferons. So we supposed that its linked with IRF8. It is expressed in non metastatic condition. VAV1 seems to be an oncogenic stress activator implicated in multiple cancer such as the lung cancer (28).

These four MRs could therefore reduce the risk of metastasis development in osteosarcoma.

The opposite roles of HOXC4 and the other four genes in the expression of genes related to metastasis phenotype are confirmed in the regulation network **Fig. 6**. In fact, HOXC4 and the four other genes regulate sometimes the same genes, and, in such cases, the regulation receives conflicting information: genes positively regulate by VAV1, IRF8 or SP140 are negatively correlated by HOXC4. (We do not observed any direct conflicts between HOXC4 and VENTX.) And, in the same time, VENTX, IRF8, VAV1 and SP140 have also common targets which they all excited, a part of them correspond to differentially expressed genes. IRF8, VENTX and SP140 share a lot of information, they are a lot of similar targets. Indeed, we saw that this three MRs are close to each other in the dendrogram. VAV1 and HOXC4 are also strongly associate according to the dendrogram. To put it in a nutshell, the inferred TN allows inferred potentials MRs implicated upstream the cascade of signalisation of osteosarcomic metastasis differentiation and proliferation. Therefore TNs enables to target new genes implicated in pathologies, and maybe target it in futur gene therapy.

But since such network are infered, we need to compare many datasets from differents hospital, country to confirm it. Moreover, the techniques used in our project is kind of old. In our data, a lot of probes haven't any gene ID, therefore, we lost around 37% of our informations. Plus, it remain some duplicated data (with no name), so it's explain why our logFC ranked see **Fig. 5** and **Fig. 7** go from 0 to 25 000. Finally, we

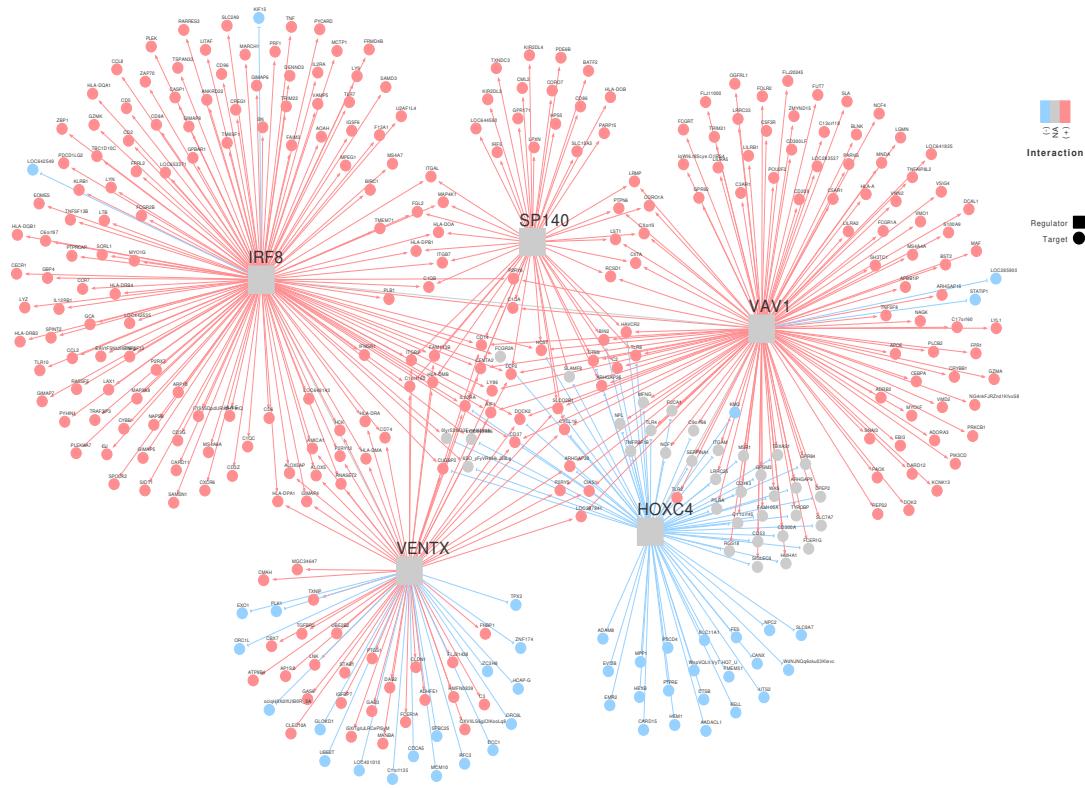


Fig. 6. Regulatory network of the top 5 MRs : MRs are represented by the largest sized squares and regulate the smaller ones (TFs targets). The colors of the edges depict the action mode of the MRs, with red bridge for an activation and a blue edge for an inhibition. Grey square are activated by certain MRs and inhibited by others

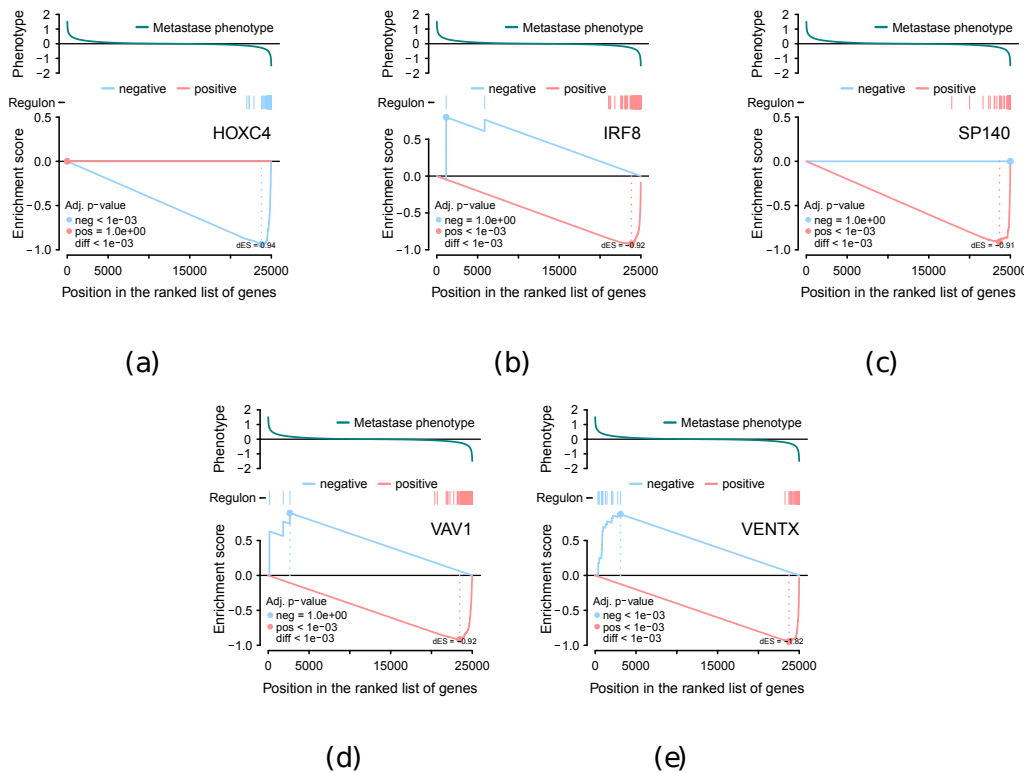


Fig. 7. Two-tailed GSEA for the top 5 MRs : Each subgraph represent the 2-tailed get set enrichment analysis for HOXC4, IRF8, SP140, VAV1 and VENTX respectively. The x-axis represent the list of gene ranked from 0 to 25 000 by their respective logFC. The upper part of the graph represent in y-axis the phenotypes ranked via the magnitude of a certain phenotypes (logFC) from -2 (no metastasis) to 2 (metastasis). The lowest part illustrate the ES with red line for regulon that are activated for a certain phenotype and blue for the inactive ones.

had to run 100 bootstrap because of the consumption of memory in our computer, we assume that it doesn't change our result, but it will be better if we do more. Maybe it will remove more instable interactions. Finally, this project highlighted 5 MRs supposedly implicated in osteosarcomatic metastasis. Because this cancer is highly deadly, it will be a welcomed future approach to focused on.

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Supporting Information (SI).

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